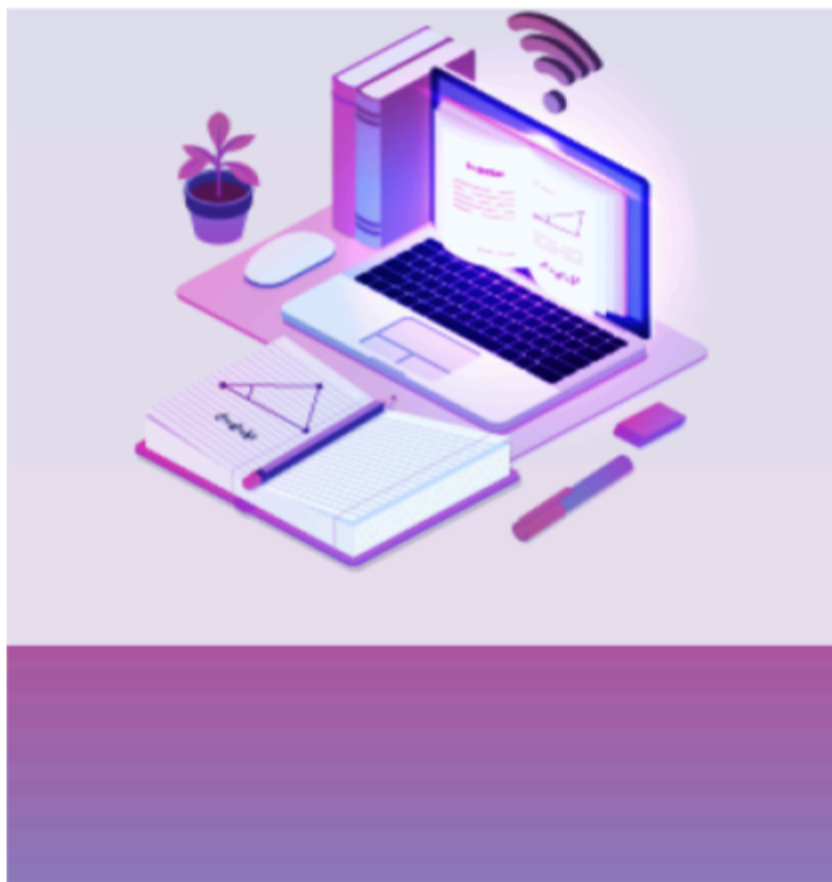


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CAT Remainder Theorem Formulas PDF

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CAT Remainder Theorems

Formulas

Fermat's little Theorem:

- Fermat's theorem is an important remainder theorem which can be used to find the remainder easily.
- Fermat's theorem states that for any integer 'a' and prime number 'p', $a^p - a$ is always divisible by 'p'.
- Also, if a is not divisible by p, i.e, if a and p are relatively prime, then $a^{(p-1)} \bmod p = 1 \bmod p$. Which means the remainder is 1.
- The second part of the theorem is very useful in solving problems.
- **Example:** when 2^{256} is divided by 17, the remainder would be _____:

Here, 7 is a prime number and 2, 17 are relatively prime.

Therefore, $2^{16} \bmod 17 = 1$.

2^{256} can be written as $(2^{16})^{16}$.

Since, $2^{16} \bmod 17 = 1$, $(2^{16})^{16} \bmod 17 = 1$.

Thus, the remainder when 2^{256} is divided by 17 is 1.

- **Example:**

Find the remainder when 3^{75} is divided by 37.

Here, 37 is a prime number. Hence, Fermat's theorem can be used. Also, 3 and 37 are relatively prime.

Therefore, $3^{36} \bmod 37 = 1$

$3^{72} \bmod 37 = (3^{36})^2 \bmod 37 = 1$

$3^{75} \bmod 37 = 3^{72} \cdot 3^3 \bmod 37 = 3^3 \bmod 37$

$27 \bmod 37$ is equal to 27.

Hence, the remainder when 3^{75} is divided by 37 is 27.



Euler's Totient:

- Euler's theorem is one of the most important remainder theorems. It is imperative to know about Euler's totient before we can use the theorem.
- Euler's totient is defined as the number of numbers less than 'n' that are co-prime to it.
- It is usually denoted as $\phi(n)$.
- The formula to find Euler's totient is

$$\phi(n) = n * \left(1 - \frac{1}{a}\right) * \left(1 - \frac{1}{b}\right) * \dots \text{where } a, b$$

are the prime factors of the numbers.

Eg: Find the number of numbers that are less than 30 and are co-prime to it.

30 can be written as $2 * 3 * 5$.

$$\phi(30) = 30 * \left(\frac{1}{2}\right) * \left(\frac{2}{3}\right) * \left(\frac{4}{5}\right) = 8$$

Therefore, 8 numbers less than 30 are co-prime to it.

→ Euler's theorem states that $a^{\phi(n)} \pmod n = 1 \pmod n$ if 'a' and 'n' are co-prime to each other.

So, if the given number 'a' and the divisor 'n' are co-prime to each other, we can use Euler's theorem.

● Example 1:

What is the remainder when 2^{256} is divided by 15?

2 and 15 are co-prime to each other. Hence, Euler's theorem can be applied.

15 can be written as 5×3 .

$$\begin{aligned} \text{Euler's totient of } 15 &= 15 \times \left(1 - \frac{1}{3}\right) \times \left(1 - \frac{1}{5}\right) \\ &= 15 \times \frac{2}{3} \times \frac{4}{5} = 8 \end{aligned}$$

Therefore, we have to try to express 256 as $8k + \text{something}$. 256 can be expressed as 8×32



We know that, $a^{\phi(n)} \pmod{n} = 1 \pmod{n}$

$$2^{8 \cdot 32} \pmod{15} = 1 \pmod{15}.$$

Therefore, 1 is the right answer.

- **Example 2:**

What are the last 2 digits of 7^{2008} ?

Finding the last 2 digits is similar to finding the remainder when the number is divided by 100.

100 and 7 are co-prime to each other. Hence, we can use Euler's theorem.

100 can be written as $2^2 * 5^2$.

Euler's totient of 100, $\phi(100)$

$$= 100 * \left(1 - \frac{1}{2}\right) * \left(1 - \frac{1}{5}\right)$$

$$= 100 * \frac{1}{2} * \frac{4}{5}$$

$$\phi(100) = 40$$

7^{2008} can be written as $7^{2000} * 7^8$

7^{2000} can be written as 7^{40*25} , Hence, 7^{2000} will yield a remainder of 1 when divided by 100

The problem is reduced to what will be the remainder when 7^8 is divided by 100

We know that $7^4 = 2401$

$$7^8 = 7^4 * 7^4 = 2401 * 2401$$

As we can clearly see, the last 2 digits will be 01.

Wilson's Theorem:

→ According to Wilson's theorem for prime number 'p', $[(p+1)!+1]$ is divisible by p.

→ In other words, $(p-1)!$ Leaves a remainder of $(p-1)$ when divided by p.

$$\text{Thus, } (p-1)! \text{ Mod } p = p-1$$



For Example:

- $4!$ When divided by 5, we get 4 as remainder.
- $6!$ When divided by 7, we get 6 as remainder.
- $10!$ When divided by 11, we get 10 as a remainder.
- If we extend Wilson's theorem further, we get an important corollary $(p-2)! \text{ Mod } p = 1$
- As from the wilson's theorem we have, $(p-1)! \text{ Mod } p = (p-1)$
- Thus, $[(p-1)(p-2)!] \text{ mod } p = (p-1)$
- This will be equal to $[(p-1) \text{ mod } p] * [(p-2)! \text{ Mod } p] = (p-1)$
- For any prime number 'p', we observe that $(p-1) \text{ mod } p = (p-1)$
- For e.g $6 \text{ mod } 7$ will be 6
- Thus, $(p-1) * [(p-2)! \text{ Mod } p] = (p-1)$

thus , for RHS to be equal to LHS,

$$(p-2)! \text{ Mod } p = 1$$

Hence, $5! \text{ Mod } 7$ will be 1 and $51! \text{ Mod } 53$ will be 1

- **Examples:**
- Q.1) What will be the remainder when $568!$ Is it divided by 569?

Solution: According to Wilson's theorem we have for prime number 'p'. $(p-1)! \text{ Mod } p = (p-1)$

In this case 569 is a prime number.

$$\text{Thus, } 568! \text{ Mod } 569 = 568$$

Hence, when $568!$ divided by 569 we get 568 as remainder.

Answer: 568



- Q.2) What will be the remainder when $225!$ is divided by 227?

Solution: We know that for the prime number 'p',
 $(p-2)! \text{ Mod } p = 1$.

In this case, 227 is a prime number.

Thus, $225! \text{ Mod } 227$ will be equal to 1.

In other words, when $225!$ is divided by 227 we get the remainder as 1.

Answer: 1

- Q.3) What will be the remainder when $15!$ is divided by 19?

Solution: 19 is a prime number.

→ From the corollary of Wilson's theorem, for prime number 'p'.

$$(p-2)! \bmod p = 1$$

$$\text{Thus, } 17! \bmod 19 = 1$$

$$[17 \cdot 16 \cdot 15!] \bmod 19 = 1$$

$$[17 \bmod 19] \cdot [16 \bmod 19] \cdot [15! \bmod 19] = 1$$

$$[-2] \cdot [-3] \cdot [15! \bmod 19] = 1$$

$$[6 \cdot 15!] \bmod 19 = 1$$

→ Multiplying both sides by 3, we get

$$[18 \cdot 15!] \bmod 19 = 3$$

$$[-1 \cdot 15!] \bmod 19 = 3$$

→ Multiplying both sides by -1, we get

$$15! \bmod 19 = -3$$

→ Remainder of '-3' when divided by 19 is the same as the remainder of '16' when divided by 19.

$$\text{Thus } 15! \bmod 19 = 16$$

Answer: 16



- Q.4) What will be the remainder when $(23!)^2$ is divided by 47?

Solution: 47 is a prime number.

→ From corollary of Wilson's theorem, for prime number 'p', $(p-1)! \mod p = -1$

Thus $46! \mod 47 = -1$

$[46 \cdot 45 \cdot 44 \cdot 43 \cdot \dots \cdot 25 \cdot 24 \cdot 23!] \mod 47 = -1$

$[(-2) \cdot (-3) \cdot (-4) \cdot \dots \cdot (-22) \cdot (-23) \cdot 23!] \mod 47 = 1$

→ We see that there are even numbers of terms from '-2' to '-23'. Thus negative signs cancel off.

→ We get

$[23! \cdot 23!] \mod 47 = 1$

Thus, $(23!)^2 \mod 47 = 1$

→ Hence, when $(23!)^2$ is divided by 47, we get 1 as a remainder.

Answer: 1

Chinese Remainder Theorem:

- The Chinese remainder theorem is useful when the divisor of any number is composite.
- Let M be a number which is divided by a divisor N .
The theorem states that if N is the divisor which can be expressed as $N = a*b$ where a and b are co-prime
- Then, $M \bmod N = ar_2x + br_1y$

Here $r_1 = M \bmod a$

And $r_2 = M \bmod b$

Here, $ax + by = 1$

- **Example 1**
- Find the remainder when 344^{237} is divided by 119
In the first look it looks difficult but if one knows the chinese remainder theorem then the question can be solved very easily.



- $119 = 17 \times 7$, so here $a = 17$ and $b = 7$

$$344^{237} \bmod 17 = 4^{237} \bmod 17 = (4 \times 16^{116}) \bmod 17 = 4 \times 1 = 4$$

Hence, we get $r_1 = 4$

Now, $344^{237} \bmod 7 = 1^{237} \bmod 7 = 1$, Hence $r_2 = 1$

We know that $M \bmod N = ar_2x + br_1y$

- Therefore, $344^{237} \bmod 119 = 17 \times 1x + 7 \times 4y = 17x + 28y$

We know that $17x + 7y = 1$

- We can see that $x = 5$ and $y = -12$ satisfies the above equation.

Hence, putting the values of x and y in equation 1, we get $344^{237} \bmod 119 = 17 \times 5 - 28 \times 12 = 85 - 336 = -251$

Converting this into positive remainder we get

$$357 - 251 = 106$$

- Hence, the required remainder is 106.
-

Example 2:

- Let's consider another example to understand is better find the remainder when 495^{2517} is divided by 78.
- In this question also, the divisor is 78 which can be written as 13×6 . So, we can use the Chinese remainder theorem in this question as well.

Let's take $a = 13$ and $b = 6$

So we can write $495^{2517} \bmod 78 = 13r_2x + 6r_1y$

$$\rightarrow r_1 = 495^{2517} \bmod 13 = 1^{2517} \bmod 13 = 1$$

$$\begin{aligned} \rightarrow r_2 &= 495^{2517} \bmod 6 = 3^{2517} \bmod 6 = (2^{2516} \bmod 2) * 3 = \\ &1 * 3 = 3 \end{aligned}$$

$$\rightarrow \text{We also know that } 13x + 6y = 1$$

$x = 1$ and $y = -2$ satisfies the above equation.

$\rightarrow$ Hence, we can obtain the remainder as

$$\begin{aligned} 495^{2517} \bmod 78 &= 13r_2x + 6r_1y = 13 * 3 * 1 - 6 * 1 * 2 \\ &= 39 - 12 = 27 \end{aligned}$$

Hence, the required answer is 27.





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